

Generalisability of the 1 bit Quantum Filter (1BQF) Beyond Track Reconstruction

A response to Reviewer 2’s “too tailored to this Hamiltonian” concern

LHCb Quantum Tracking working note

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Abstract

The reviewer asks whether the spectral-filter trick at the heart of our quantum track-reconstruction pipeline – which we will refer to as the *one-bit quantum filter* (1BQF) – is specific to the LHCb VELO Hopfield Hamiltonian, or whether it represents a more general computational primitive. This note argues, with citations, that 1BQF is a *problem-class-level* primitive: it is the minimal quantum linear-system (HHL-type) circuit whenever (i) the system matrix admits a *bimodal* spectrum with one cluster of physically irrelevant “noise” eigenvalues, (ii) the right-hand side preparation is uniform or otherwise gate-cheap, and (iii) the quantity of interest is a low-weight expectation value of the solution, not the full solution vector. We identify families of problems where these three conditions hold by construction (associative memory recall, graph-Laplacian / Hopfield-QUBO optimisation, kernel-method classifiers, fitting with quadratic regularisers, certain PDE inverse problems, image template-matching), and delimit where they do not (dense linear regression with arbitrary right hand sides, problems requiring full state tomography, ill-conditioned systems with continuous spectra).

1 What the 1 bit Quantum Filter actually is

In the published track-reconstruction work the quantum solver receives a Hopfield-Hamiltonian linear system $\mathbf{A} \mathbf{x} = \mathbf{b}$ constructed from the segment graph of a VELO event, with $\mathbf{A}$ a real symmetric positive-definite matrix of dimension N equal to the number of admissible doublet/segment candidates. The key spectral observation – already implicit in the original Hopfield/Stimpfl-Abele formulation [1, 2, 3] – is that $\mathbf{A}$ has a *bimodal* spectrum:

- a small cluster of “signal” eigenvalues $\{\lambda_s\}$, whose eigenvectors are supported on the segments belonging to true tracks;
- a large cluster of “noise” eigenvalues, concentrated around a single, analytically known value $\lambda_n \approx \gamma$ (the conflict-penalty coupling), whose eigenvectors are the Hopfield “basins” of incompatible-segment configurations.

1BQF is the HHL pipeline of [4] reduced to a *single* clock (phase-estimation) qubit, $n_t = 1$. Because the phase register has only two computational-basis states, the controlled inverse rotation can resolve only *two* eigenvalue bins. This is exactly the resolution required: one bin captures the noise cluster and its amplitude is rejected via the ancilla post-selection, the other bin captures the signal cluster. Schematically, with $\Pi_{\text{sig}} = \sum_{\lambda \in \{\lambda_s\}} |\lambda\rangle\langle\lambda|$ and $\Pi_{\text{noise}} = \mathbb{I} - \Pi_{\text{sig}}$,

$$|\psi_{\text{post}}\rangle \propto \Pi_{\text{sig}} |\mathbf{b}\rangle + \mathcal{O}(\sqrt{p_{\text{leak}}}) \Pi_{\text{noise}} |\mathbf{b}\rangle, \quad (1)$$

where p_{leak} is set by the spectral gap $\Delta = |\lambda_n - \lambda_s|$ and the QPE resolution. The final answer is read out as the expectation value of an $\mathcal{O}(1)$ -weight observable (in our case the which-segment-is-active projector on a chosen qubit), not as a full amplitude vector.

The three structural ingredients that make this work are therefore:

- (S1) **Bimodal spectrum** with one analytically known cluster eigenvalue λ_n that is irrelevant to the answer.
- (S2) **Gate-cheap right-hand side:** $|\mathbf{b}\rangle$ preparable without QRAM (in our case $|\mathbf{b}\rangle = H^{\otimes n_s}|0\rangle$, the uniform superposition).
- (S3) **Low-weight observable:** the answer is $\langle\psi|O|\psi\rangle$ with O supported on $\mathcal{O}(1)$ qubits, not the amplitudes of $|\psi\rangle$ themselves.

None of (S1)–(S3) refers to tracking, to LHCb, or even to physics. Any problem that admits this triple is an 1BQF problem.

2 Why 1BQF is structurally transferable

1BQF is best understood as a special case of the broader family of *quantum eigenvalue / singular-value transformations* (QSVT) [5, 6], in which a polynomial $P(\mathbf{A})$ of the system matrix is implemented coherently, and matrix inversion is recovered by choosing $P(\lambda) \approx 1/\lambda$. The full QSVT construction is overkill if one only needs the projector $P(\lambda) \approx \mathbf{1}_{\lambda \in [\lambda_s^-, \lambda_s^+]}$ that separates two clusters: a single bit of phase resolution is sufficient by Kitaev’s iterative phase-estimation argument [7, 8]. In other words 1BQF is the *cheapest non-trivial QSVT* that still solves a useful problem; it is interesting precisely because so many useful problems satisfy (S1)–(S3) and therefore do not require the full polynomial machinery.

The transferability statement is then:

Whenever a linear-algebraic task can be re-cast so that its system matrix has a bimodal spectrum, its right-hand side is gate-cheap, and its answer is a low-weight observable, the same 1BQF circuit – up to a problem-specific Hamiltonian-simulation block for $e^{-i\mathbf{A}t}$ – solves it with the same $\mathcal{O}(\sqrt{N} \log N)$ gate budget reported in our paper.

This argument is structural: it reduces the question “can your method generalise?” to the question “does my problem have a bimodal spectrum?”. The latter is a spectral-graph-theory question that can be answered before any quantum circuit is written, by classical sparse-eigenvalue analysis (in our work this is done by ARPACK [9]).

3 Families of problems that satisfy (S1)–(S3)

We list, in decreasing order of structural similarity to our tracking application, the problem families to which 1BQF transfers. In each case we name the source matrix, the “noise” eigenvalue, the natural observable, and a representative reference.

F1. Hopfield-style associative memory and pattern recognition

The original Hopfield network [1] stores patterns as fixed points of a quadratic energy $E(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^\top \mathbf{W}\mathbf{x}$, recall is linear inversion in the basin of a stored pattern, and the spurious-pattern spectrum is concentrated on a single loading-fraction-dependent eigenvalue [10]. Quantum associative-memory proposals [11, 12] are exactly in this regime; 1BQF provides the missing inversion/recall step with single-clock-qubit cost. Track reconstruction is one instance of this family: any combinatorial pattern-completion problem with a Hopfield encoder qualifies (e.g. content-addressable memory, image inpainting on small quantum-tractable patches).

F2. Graph-Laplacian / spectral-clustering linear systems

For a graph with combinatorial Laplacian $\mathbf{L}$, problems such as seeded community detection, electrical-network effective-resistance queries, and PageRank-with-restart are linear systems $(\mathbf{L} + \alpha\mathbf{1})\mathbf{x} = \mathbf{b}$. The Laplacian of a graph with k well-separated communities has k small eigenvalues and a clearly

separated bulk [13]; the bulk eigenvalue acts as λ_n and the answer (community label of a query node, effective resistance, etc.) is a low-weight observable. QSVT-based approaches to such problems already exist [14, 15] and 1BQF is the natural single-clock-qubit specialisation.

F3. Quadratic unconstrained binary optimisation (QUBO) with structured penalties

Our Hamiltonian is a QUBO with a large conflict penalty γ that pins the “noise” eigenvalue. This is the standard pattern in QUBO/Ising encodings of constrained combinatorial problems [16]: max-cut, graph colouring, set cover, vertex cover, and maximum independent set all have penalty matrices whose spectrum clusters around the penalty strength. When the question of interest is “is variable i active in the dominant low-energy mode?” (a low-weight observable on a single qubit), 1BQF returns the answer directly without sampling the full ground state. This makes 1BQF complementary to QAOA [17] and quantum annealing [18]: instead of preparing the ground state, it projects onto the low-energy subspace and reads a marginal.

F4. Kernel methods and least-squares classifiers (LS-SVM, ridge regression, Gaussian-process regression)

A regularised kernel system $(\mathbf{K} + \lambda \mathbf{1})\boldsymbol{\alpha} = \mathbf{y}$ has a spectrum dominated by a few large kernel eigenvalues plus a long tail near the regulariser λ [19]; the regulariser pins λ_n . When the test prediction is needed only for a few new query points – a low-weight observable on the output register – this is exactly an 1BQF problem. Quantum kernel/SVM approaches based on full HHL [20] can be reduced to single-clock-qubit cost whenever the kernel is well-conditioned and the prediction observable is local, which is the setting most often encountered in HEP fast-inference applications.

F5. Linear PDEs with a known spectral gap

Discretised elliptic PDEs $-\nabla \cdot (c \nabla u) = f$ on a quasi-uniform mesh produce a stiffness matrix whose eigenvalues separate into a small set tied to global modes and a bulk tied to the local discretisation scale; this gap is the basis of multigrid preconditioning. Quantum PDE solvers [21, 15, 22] that read out only a local functional of u (a flux, an integrated source term) are 1BQF-compatible. Track extrapolation through a piecewise-uniform field map – a directly LHCb-relevant PDE/ODE-inverse-problem – is in this class.

F6. Template matching and constrained image segmentation

When the answer asked of an image is binary – “does pattern T appear at position $\mathbf{p}$?” – the underlying linear problem is again a quadratic projection onto a known template subspace whose spectrum is bimodal (template / non-template). This generalises the “did this pattern of hits form a track?” question of our paper. Quantum template-matching circuits already use spectral-projection ideas [23] and admit a one-bit reduction.

The list is not exhaustive but is sufficient to answer Reviewer 2’s question: the spectral-filter trick is not Hamiltonian-specific; it works whenever the spectrum has the right shape and the quantity of interest is a low-weight observable.

4 Where 1BQF does *not* transfer

For completeness we also state the boundaries of the method, since this is also part of an honest transferability discussion.

- **Continuous spectra without a gap.** Dense linear regression with an arbitrary covariance matrix, or a stiff PDE with no clear discretisation gap, will spread amplitude across the single phase bit and incur $\mathcal{O}(1)$ filter leakage. Multi-bit phase estimation or full QSVT is then required.

- **Tomography-bound answers.** If the answer is the full solution vector $\mathbf{x}$ rather than $\langle \mathbf{x} | O | \mathbf{x} \rangle$ for a low-weight O , the readout cost is $\mathcal{O}(N)$ regardless of the inversion step [24], and any quantum speed up is erased – this is precisely the issue Reviewer 2 raised with reference to [25] and we addressed in our Major-1 reply.
- **Expensive state preparation.** If $|\mathbf{b}\rangle$ cannot be prepared without QRAM, condition (S2) fails and the speed-up is again lost. Our tracking encoder happens to satisfy (S2) by Hadamards alone; transferred problems should be checked case by case.
- **Non-Hermitian or unbounded operators.** 1BQF requires Hamiltonian simulation $e^{-i\mathbf{A}t}$. Non-normal matrices (e.g. certain advection-dominated PDE discretisations) need block encoding via QSVT instead.

5 Recommendation for the manuscript

We propose adding to the manuscript a single short subsection, “*Generality of the spectral-filter mechanism*”, that: (i) states the three structural ingredients (S1)–(S3); (ii) cites the QSVT/IPE generalisation [5, 6, 7]; (iii) lists by name two or three of the families F1–F6 above as illustrative examples (associative memory, QUBO with structured penalties, regularised kernel methods); (iv) explicitly states the limits of Section 4.

This addresses Reviewer 2’s “too tailored” concern without overclaiming generality: the method is not a universal quantum solver, but the structural conditions under which it works are common, well-defined, and verifiable in advance by classical spectral analysis.

6 Proposed reviewer reply

The techniques presented in this work were developed to make TrackHHL executable on quantum hardware by reducing qubit count, two-qubit gate count, and circuit depth – the standard bottlenecks of HHL-based pipelines. As the reviewer correctly notes, the optimizations exploit the geometrical structure of the tracking problem and the numerical features of our quadratic Hamiltonian, and this degree of problem-specificity is typical of HHL applications, since HHL is not currently regarded as a general-purpose replacement for classical matrix inversion. We nevertheless believe that the methods transfer to a well-defined family of problems, and we make the criterion for transferability mathematical rather than application-driven.

Concretely, our one-bit quantum filter (1BQF) solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ on the subspace where it is informative, provided the input data $(\mathbf{A}, \mathbf{b}, O)$ satisfy the following structural conditions:

- (M1) $\mathbf{A} \in \mathbb{R}^{N \times N}$ is symmetric positive semi-definite, s -sparse with $s = \mathcal{O}(\text{polylog } N)$, and admits an efficient Hamiltonian-simulation oracle for $e^{-i\mathbf{A}t}$ (in our case, the sparse-segment encoder of Sec. III).
- (M2) The spectrum of $\mathbf{A}$ decomposes as $\sigma(\mathbf{A}) = \sigma_{\text{sig}} \cup \sigma_{\text{noise}}$ with $\sigma_{\text{noise}} \subseteq [\lambda_n - \delta, \lambda_n + \delta]$ around an analytically known eigenvalue λ_n , and a finite spectral gap $\Delta = \text{dist}(\sigma_{\text{sig}}, \sigma_{\text{noise}}) > 0$ with $\Delta \gg \delta$. The condition number on σ_{sig} is bounded by a problem-independent constant $\kappa_{\text{sig}} = \mathcal{O}(1)$.
- (M3) $|\mathbf{b}\rangle$ is preparable by an $\mathcal{O}(\text{polylog } N)$ -depth circuit without QRAM (in our case $|\mathbf{b}\rangle = H^{\otimes n_s} |0\rangle$, the uniform superposition).
- (M4) The quantity of interest is $\langle \mathbf{x} | O | \mathbf{x} \rangle / \|\mathbf{x}\|^2$ for an observable O supported on $\mathcal{O}(1)$ qubits, e.g. a single-mode population $O = |j\rangle\langle j|$ or a low-rank projector $O = \sum_{j \in S} |j\rangle\langle j|$ with $|S|$ bounded independently of N .

Whenever (M1)–(M4) hold, the same circuit applies up to a problem-specific Hamiltonian-simulation block, with the same $\mathcal{O}(\sqrt{N} \log N)$ gate budget reported in our paper. This is the regime considered by Rebentrost et al. [26], from which TrackHHL is inspired: optimization of a Hopfield network

is recast as a sparse linear problem solved by HHL, with an exponential advantage provided that efficient Hamiltonian simulation and readout subroutines are available, and that the linear system can be suitably regularized. Our 1BQF construction sharpens this picture: *regularization* in their language is exactly the noise-eigenvalue pinning of (M2), and *efficient readout* is exactly the low-weight observable of (M4). When both hold, the multi-qubit phase register of standard HHL collapses to a single clock qubit, as the only spectral information that has to be resolved is the binary signal/noise label.

The structural conditions (M1)–(M4) define a family of problems mathematically rather than by domain. Within this family one finds, in addition to the RNA-reconstruction example outlined by Rebentrost et al.: Hopfield-style associative memories and pattern denoising, where $\mathbf{A}$ is a Hebbian coupling matrix whose spurious-pattern spectrum concentrates on a known loading-fraction-dependent eigenvalue [1, 10]; QUBO/Ising encodings of constrained combinatorial problems whose conflict-penalty term sets λ_n by construction (max-cut, graph colouring, vertex cover, set cover, MIS; see [16]), where 1BQF returns a marginal of the low-energy subspace and is complementary to QAOA [17] and quantum annealing [18]; graph-Laplacian linear systems $(\mathbf{L} + \alpha\mathbf{1})\mathbf{x} = \mathbf{b}$ on graphs with k well-separated communities, whose Laplacian spectrum has k small eigenvalues separated from a clearly defined bulk [13, 14, 15], with applications in seeded community detection and effective-resistance queries; regularized kernel systems $(\mathbf{K} + \lambda\mathbf{1})\boldsymbol{\alpha} = \mathbf{y}$ in least-squares SVMs, ridge regression, and Gaussian-process regression [19, 20], where the regularizer pins $\lambda_n = \lambda$ and the prediction at a test point is by construction a local observable on the output register; and discretized linear PDEs $\mathbf{L}_h\mathbf{u} = \mathbf{f}$ with a clear spectral gap between global and discretization-scale modes [21, 22], provided only a local functional of $\mathbf{u}$ is read out – a class that includes track extrapolation through piecewise-uniform field maps.

Conversely, (M1)–(M4) are sharp, and the method does *not* transfer to problems with continuous, gapless spectra ($\Delta \rightarrow 0$, where filter leakage becomes $\mathcal{O}(1)$); to tasks requiring full-state tomography (where the $\mathcal{O}(N)$ readout cost erases any speed-up, as discussed in our reply to Reviewer 2 Major-1 [24, 25]); to settings where state preparation requires QRAM and (M3) fails; or to non-normal $\mathbf{A}$, for which Hamiltonian simulation must be replaced by block encoding via QSVT [5, 6]. We expect that, as in the present work, application to any of these problems will still require adaptation to the specific structure and constraints of the system at hand, but the conditions under which the adaptation is worthwhile are well-defined and can be verified classically – by sparse spectral analysis of $\mathbf{A}$ – before any quantum circuit is written.

References

- [1] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982.
- [2] G. Stimpfl-Abele and L. Garrido. Fast track finding with neural networks. *Computer Physics Communications*, 64(1):46–56, 1991.
- [3] B. Denby. Neural networks and cellular automata in experimental high energy physics. *Computer Physics Communications*, 49(3):429–448, 1988.
- [4] A. W. Harrow, A. Hassidim, and S. Lloyd. Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15):150502, 2009.
- [5] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC ’19)*, pages 193–204, 2019.
- [6] J. M. Martyn, Z. M. Rossi, A. K. Tan, and I. L. Chuang. Grand unification of quantum algorithms. *PRX Quantum*, 2(4):040203, 2021.

- [7] A. Yu. Kitaev. Quantum measurements and the Abelian stabilizer problem, 1995.
- [8] B. L. Higgins, D. W. Berry, S. D. Bartlett, H. M. Wiseman, and G. J. Pryde. Entanglement-free heisenberg-limited phase estimation. *Nature*, 450:393–396, 2007.
- [9] R. B. Lehoucq, D. C. Sorensen, and C. Yang. ARPACK users’ guide: Solution of large-scale eigenvalue problems with implicitly restarted arnoldi methods. 1998.
- [10] D. J. Amit, H. Gutfreund, and H. Sompolinsky. Spin-glass models of neural networks. *Physical Review A*, 32(2):1007–1018, 1985.
- [11] D. Ventura and T. Martinez. Quantum associative memory. *Information Sciences*, 124(1–4):273–296, 2000.
- [12] C. A. Trugenberger. Probabilistic quantum memories. *Physical Review Letters*, 87(6):067901, 2001.
- [13] U. von Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416, 2007.
- [14] S. Apers and R. de Wolf. Quantum speedup for graph sparsification, cut approximation and Laplacian solving. *SIAM Journal on Computing*, 51(6):1703–1742, 2022.
- [15] A. M. Childs, R. Kothari, and R. D. Somma. Quantum algorithm for systems of linear equations with exponentially improved dependence on precision. *SIAM Journal on Computing*, 46(6):1920–1950, 2017.
- [16] A. Lucas. Ising formulations of many NP problems. *Frontiers in Physics*, 2:5, 2014.
- [17] E. Farhi, J. Goldstone, and S. Gutmann. A quantum approximate optimization algorithm, 2014.
- [18] T. Kadowaki and H. Nishimori. Quantum annealing in the transverse Ising model. *Physical Review E*, 58(5):5355–5363, 1998.
- [19] C. K. I. Williams and C. E. Rasmussen. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [20] P. Rebentrost, M. Mohseni, and S. Lloyd. Quantum support vector machine for big data classification. *Physical Review Letters*, 113(13):130503, 2014.
- [21] D. W. Berry. High-order quantum algorithm for solving linear differential equations. *Journal of Physics A: Mathematical and Theoretical*, 47(10):105301, 2014.
- [22] D. W. Berry, A. M. Childs, A. Ostrander, and G. Wang. Quantum algorithm for linear differential equations with exponentially improved dependence on precision. *Communications in Mathematical Physics*, 356:1057–1081, 2017.
- [23] X.-D. Cai et al. Entanglement-based machine learning on a quantum computer, 2015. [VERIFY: replace with definitive quantum template-matching / pattern-recognition reference before submission].
- [24] S. Aaronson. Read the fine print. *Nature Physics*, 11:291–293, 2015.
- [25] E. Tang. A quantum-inspired classical algorithm for recommendation systems. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC ’19)*, pages 217–228, 2019.
- [26] P. Rebentrost, T. R. Bromley, C. Weedbrook, and S. Lloyd. Quantum Hopfield neural network. *Physical Review A*, 98(4):042308, 2018.